Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 8

Roll No : 324

Name : Aditya Chauhan

01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}
```

In [1]:
```
a=set(map(int,input("enter data:").split()))
b=set(map(int,input("enter data:").split()))

print(f"Union: {a.union(b)} \n intersection: {a.intersection(b)} \n Difference: {a.
```

```
Union: {1, 2, 3, 4, 5, 6}
 intersection: {3, 4}
 Difference: {1, 2}
```

02) WAP to Find elements common to all three given sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {2, 3, 5}
C = {2, 3, 6}
```

Sample Output:

```
{2, 3}
```

In [2]:
```python
a=set(map(int,input("enter data:").split()))
b=set(map(int,input("enter data:").split()))
c=set(map(int,input("enter data:").split()))

print(a.intersection(b.intersection(c)))
```

```
{2, 3}
```

03) WAP to remove duplicate elements from a list while preserving unique values.

Sample Input:

```
[10, 20, 10, 30, 20, 40]
```

Sample Output:

```
{10, 20, 30, 40}
```

In [6]:
```python
a=list(map(int,input("enter data:").split()))
b=set()

for i in a:
    if i not in b:
        b.add(i)
print(b)
```

```
{40, 10, 20, 30}
```

04) WAP to count how many elements are common between two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
2
```

In [5]:
```
a=set(map(int,input("enter data:").split()))
b=set(map(int,input("enter data:").split()))

ans=a.intersection(b)

print(len(ans))
```

```
2
```

05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.

Sample Input:

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

In [19]:
```
data=input("Enter dict1 key-value pairs: ").split()
dict1={int(data[i]): int(data[i+1]) for i in range(0, len(data), 2)}
data=input("Enter dict2 key-value pairs: ").split()
dict2={int(data[i]): int(data[i+1]) for i in range(0, len(data), 2)}
dict3=dict1.copy()

for k1,v1 in dict2.items():
    if k1 in dict3:
        dict3[k1]+=v1
    else:
        dict3[k1]=v1
print(dict3)
```

```
{2: 60, 3: 80, 4: 60, 1: 10}
```

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

In [21]:
```python
data=input("Enter dict1 key-value pairs: ").split()
dict1={data[i]: int(data[i+1]) for i in range(0, len(data), 2)}
items=list(dict1.items())
for i in range(len(items)):
    for j in range(i+1,len(items)):
        if items[i][1]>items[j][1]:
            items[i],items[j]=items[j],items[i]
dict2=dict(items)
print(dict2)
```

```
[('chemistry', 60), ('math', 70), ('physics', 85)]
{'chemistry': 60, 'math': 70, 'physics': 85}
```

07) WAP to find the keys having maximum and minimum values in a dictionary.

Sample Input:

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

```
Max Value Key: B
Min Value Key: C
```

In [24]:
```python
data=input("Enter dict1 key-value pairs: ").split()
dict1={data[i]: int(data[i+1]) for i in range(0, len(data), 2)}
mn=min(dict1.values())
mx=max(dict1.values())
for k,v in dict1.items():
    if v==mn:
        print(f"min value key: {k}")
    elif v==mx:
        print(f"max value key: {k}")
```

```
max value key: b
min value key: c
```

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.

Sample Input:

```
N = 5
```

Sample Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

In [27]:
```python
n=int(input("enter value of n:"))
d=dict()
for i in range(1,n+1):
    d[i]=i**2
print(d)
```

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

09) WAP to count the frequency of each character in a string using a dictionary.

Sample Input:

```
programming
```

Sample Output:

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

In [30]:
```python
inp=input("Enter string: ")
d=dict()
for i in inp:
    if i in d:
        d[i]+=1
    else:
        d[i]=1
print(d)
```

```
{'p': 1, 'r': 2, 'o': 1, 'g': 4, 'a': 2, 'm': 3, 'i': 3, 'n': 3}
```

10) WAP to swap keys and values of a dictionary.

Sample Input:

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

In [33]:
```
data=input("Enter dict1 key-value pairs: ").split()
dict1={int(data[i]): data[i+1] for i in range(0, len(data), 2)}
dict2=dict()
for k,v in dict1.items():
    dict2[v]=k
print(dict2)
```

```
{'a': 1, 'b': 2, 'c': 3}
```

11) WAP to find the student with the highest total marks, given student names and their marks.

Sample Input:

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

```
Topper: Amit
```

In [31]:
```
data=input("Enter student names followed by their marks: ").split()
dict1={}
i=0
while i<len(data):
    name=data[i]
    marks=[]
    i+=1
    while i<len(data) and data[i].isdigit():
        marks.append(int(data[i]))
        i+=1
    dict1[name]=marks
topper=''
highest_total=0

for name, marks in dict1.items():
    total=sum(marks)
    if total>highest_total:
        highest_total=total
        topper=name
print("Topper:", topper)
```

```
Topper: amit
```

12) WAP to extract keys whose values are even numbers.

Sample Input:

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

```
[2, 4]
```

In [34]:
```python
data=input("Enter dict1 key-value pairs: ").split()
dict1={int(data[i]): int(data[i+1]) for i in range(0, len(data), 2)}
even_keys=[k for k, v in dict1.items() if v % 2 == 0]
print(even_keys)
```

```
[2, 4]
```

13) WAP to Find common keys between two dictionaries.

Sample Input:

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

```
[2, 3]
```

In [1]:
```python
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}

ans = []
for i,j in Dict1.items():
    if i in Dict2:
        ans.append(i)
print(ans)
```

```
[2, 3]
```

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
```

Sample Output - 1:

```
8
```

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'
```

Sample Output - 2:

```
Key Not Found
```

In [3]:
```python
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'

if key not in dict1:
    print("Key Not Found")
else:
    print(dict1[key])
```

```
8
```

15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

In [9]:
```python
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
ans = {students[i]:marks[i] for i in range(len(students)) if marks[i]>=75}
print(ans)
```

```
{'Amit': 78, 'Neha': 85}
```

In [ ]: